County-Level Economic Factors and High School Graduation Rates in Washington State

Authors: Louis Han, Yu Chen

1. Summary of Questions and Results

1. **How does the median household income of a geographical area affect high school graduation rates?**
   Our analysis found a weak positive relationship between county median household income and high school graduation rate. Counties with higher median household income generally tended to have slightly higher graduation rates, although the relationship was not strong and included several outliers.

2. **How does a county's high school graduation rate correlate with the percentage of families in poverty with youth under 18?**
   Our analysis found a negative relationship between child poverty percentage and graduation rate. Counties with higher child poverty percentages generally tended to have lower graduation rates, and this trend appeared more visually apparent than the income relationship.

3. **Can we predict graduation rates from socioeconomic variables such as income and poverty levels?**
   We found that county-level socioeconomic variables provided only limited predictive power for graduation rate. Among the models we tested, linear regression performed better than random forest, but both models had low explanatory power overall.

2. Motivation

High school graduation is often treated as a broad indicator of educational success, but it is also closely connected to the economic conditions in which students grow up. Access to stable income, housing, nutrition, and other resources can shape educational opportunity long before students reach graduation. Because of this, graduation rate may reflect not only school performance, but also the broader social and economic environment of a county.

Although it is commonly assumed that wealthier counties have better educational outcomes and that poverty is associated with weaker outcomes, these ideas are often discussed in general terms rather than tested directly with recent public data. In this project, we use publicly available county-level data from Washington State to study how graduation rates relate to median

household income and child poverty. We also examine whether these socioeconomic measures can be used to predict graduation rates.

Our goal is to build a clear, reproducible analysis that connects economic environment to educational outcomes at the county level. By combining education and socioeconomic datasets, we aim to better understand which patterns are supported by the data and where those relationships remain limited or incomplete.

3. Dataset

- **Dataset A: U.S. Census Bureau 2024 SAIPE County Estimates**
  - **Source:** SAIPE State and County Estimates for 2024
  - **Link:** https://www.census.gov/data/datasets/2024/demo/saipe/2024-state-and-county.html
  - **Description:** This dataset provides 2024 socioeconomic estimates for counties across the United States. Relevant variables for this project include median household income, poverty estimates for all ages, poverty percentages for all ages, child poverty estimates for ages 0–17, and child poverty percentages for ages 0–17. These variables were used to represent the economic environment of each county.
  - **Access Instructions:** 1. Navigate to the link. 2. Under the "2024" section, download the file titled "US and All States and Counties."
- **Dataset B: Washington OSPI Report Card Graduation 2024-25**
  - **Source:** Report Card Graduation 2024-25 - Catalog
  - **Link:** https://catalog.data.gov/dataset/report-card-graduation-2024-25
  - **Description:** This dataset contains graduation-related outcomes for the 2024–25 school year in Washington State. It includes records at multiple organizational levels and for multiple student groups. For this project, we filtered the data to county-level observations for all students and used graduation rate as the main educational outcome variable.
  - **Access Instructions:** 1. Visit the Data.gov link. 2. Scroll to "Downloads and Resources." 3. Click "Comma Separated Values File Download."

After cleaning and filtering, the two datasets were merged using **County** as the shared identifier. The final merged dataset contains county-level observations for Washington State and includes both graduation outcomes and socioeconomic indicators.

4. Method

Our analysis was completed in three main stages: data preparation, exploratory data analysis, and predictive modeling.

First, we cleaned and filtered the Washington OSPI graduation dataset to keep only county-level observations for **All Students**. Because the OSPI dataset includes multiple cohort definitions,

we used the **Seven Year** cohort for the final merged dataset in order to retain a broader county-level view of graduation outcomes. We converted the graduation rate column to numeric form and handled missing values programmatically.

Second, we cleaned the U.S. Census SAIPE dataset and selected the socioeconomic variables relevant to our research questions, including **MedianHouseholdIncome**, **PovertyPercent_AllAges**, **PovertyEstimate_AllAges**, **ChildPovertyPercent_0_17**, and **ChildPovertyEstimate_0_17**. We then merged the OSPI and SAIPE datasets using **County** as the common identifier to create a county-level dataset for Washington State.

Third, we conducted exploratory data analysis to understand the distribution of the variables and their relationships. We created histograms to examine the distributions of graduation rate, median household income, and child poverty percentage. We also created scatterplots with fitted regression lines to study the relationships between graduation rate and median household income, and between graduation rate and child poverty percentage.

Finally, to address the prediction question, we built two regression models using the merged county-level dataset. We used **GraduationRate** as the target variable and used the county-level socioeconomic variables as predictors. We compared a **Linear Regression** model and a **Random Forest Regressor** using mean squared error (MSE) and R^2 as evaluation metrics. This allowed us to assess both whether socioeconomic variables could be used to predict graduation rate and which model performed better on this dataset.

5. EDA

Exploratory data analysis helped us understand both the structure of the original education dataset and the properties of the final merged county-level dataset used in our analysis.

We first examined the Washington OSPI graduation dataset before merging. This dataset was large and detailed, with records at multiple organizational levels and for multiple student groups and cohort definitions. Because our project focuses on county-level educational outcomes, we filtered the dataset to county-level observations for All Students. We also explored the distribution of graduation rate across cohorts. The distribution plot for graduation rate showed that most county-level graduation rates were concentrated in the upper range, while a smaller number of counties had notably lower values.

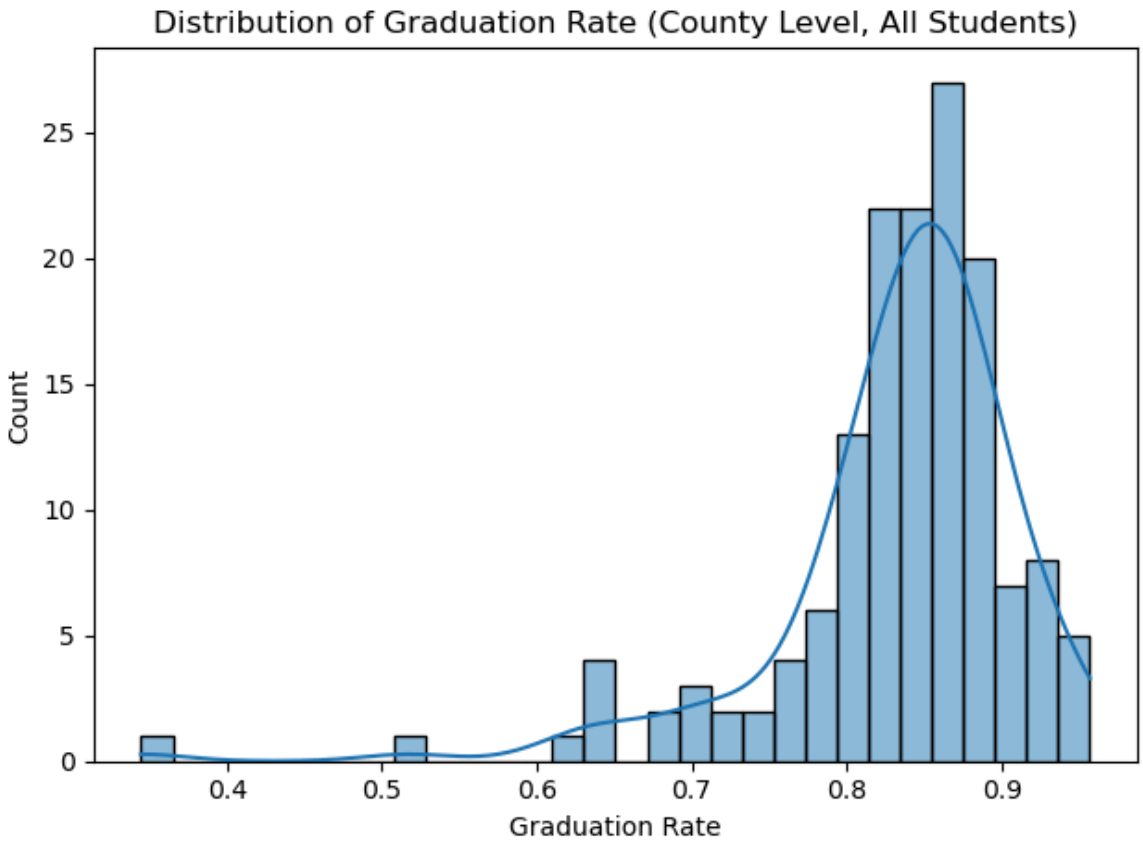

Figure 1. Distribution of county-level graduation rates for all students in Washington State. Most counties cluster in the higher graduation-rate range, with a smaller number of lower-performing outliers.

This suggested that graduation outcomes were generally high across Washington counties, but that some counties stood out as lower-performing outliers.

We also examined graduation rate by cohort using a boxplot. This comparison showed that graduation rates were fairly similar across the Four-Year, Five-Year, Six-Year, and Seven-Year cohorts, although the median graduation rate increased slightly as the cohort window expanded.

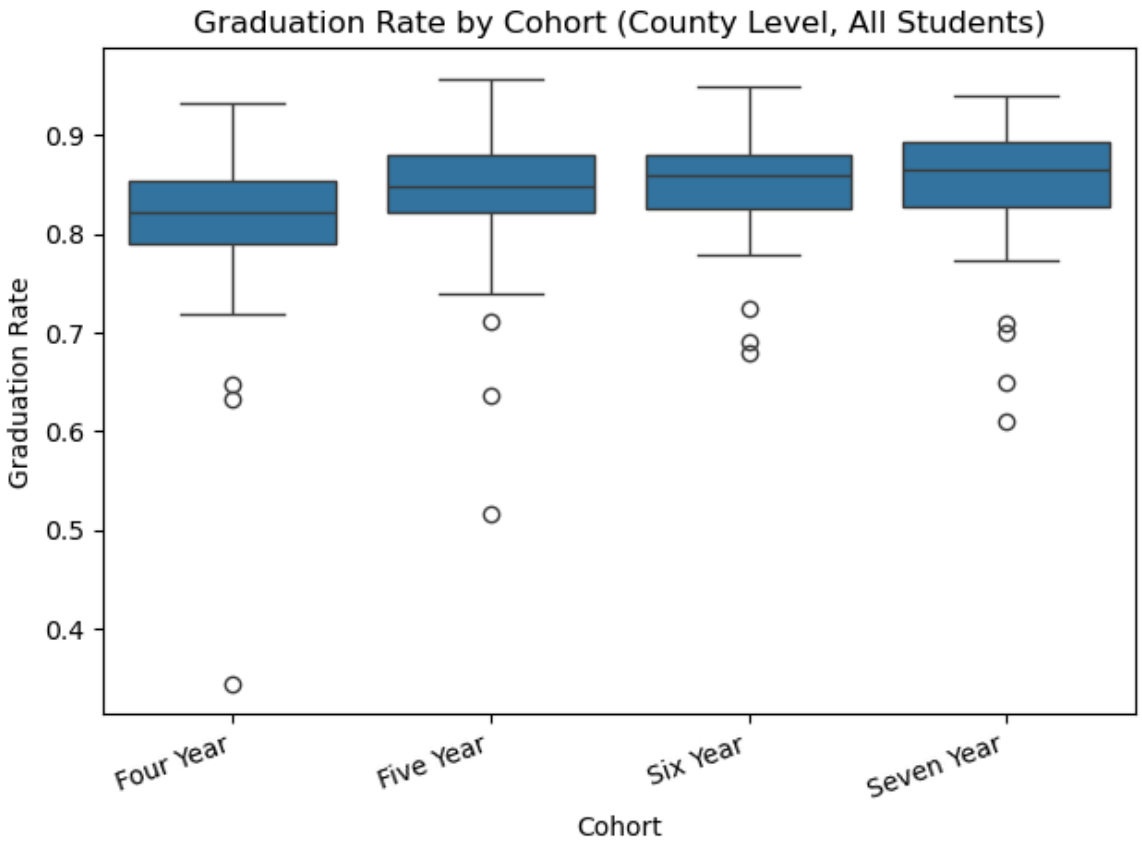

Figure 2. Graduation rate by cohort at the county level. Median graduation rate increases slightly as the cohort window expands from four years to seven years.

This pattern is expected because allowing more years gives students more time to complete graduation requirements. Because the Seven-Year cohort provides the broadest view of county graduation outcomes, we selected it for the final merged dataset.

After cleaning and merging the OSPI and SAIPE datasets, we created a county-level dataset for Washington State containing 39 rows and 10 columns, where each row represented one county. We then examined the distributions of the merged variables. Median household income showed moderate variation across counties, with some higher-income counties standing out from the rest of the state. Child poverty percentage also varied across counties, suggesting meaningful differences in county-level economic conditions.

We also used a geographic visualization of graduation rate to understand the spatial distribution of educational outcomes across Washington State.

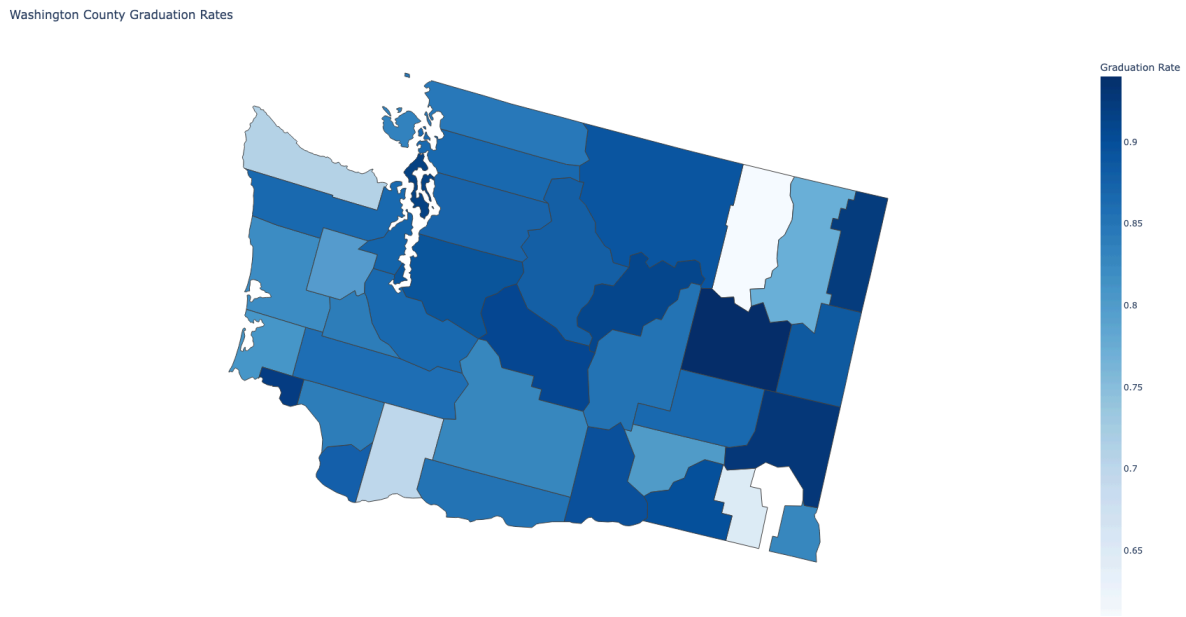

Figure 3. Geographic distribution of county graduation rates in Washington State. Graduation outcomes vary across counties rather than being evenly distributed statewide.

This map showed that graduation rates were not evenly distributed geographically. Some counties had visibly lower graduation outcomes than others, reinforcing the idea that county-level context may matter when studying educational attainment. Similarly, a geographic map of median household income showed strong regional variation, particularly with several western counties displaying noticeably higher income levels than much of the rest of the state.

Overall, the EDA stage helped us identify three important patterns that shaped the rest of the project. First, graduation rate varied enough across counties to support a meaningful county-level analysis. Second, both income and child poverty showed substantial geographic and numeric variation. Third, the original OSPI dataset required filtering and simplification before it could be meaningfully combined with the socioeconomic data. These findings motivated the final relationship analysis and predictive modeling presented in the following sections.

6. Results

6.1 Research Question 1: How does the median household income of a geographical area affect high school graduation rates?

To answer this question, we examined the relationship between county median household income and county graduation rate using a scatterplot with a fitted regression line.

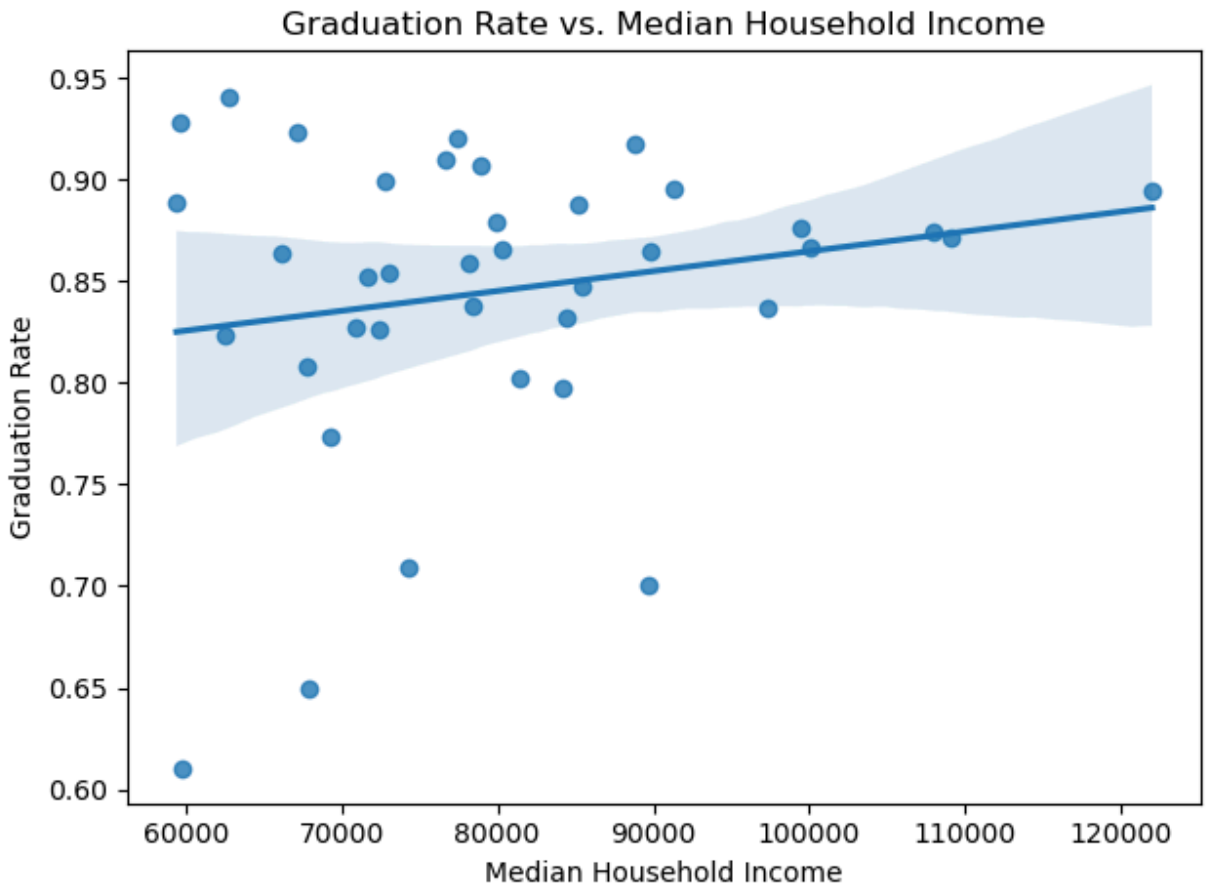

Figure 4. Relationship between county median household income and graduation rate. The fitted line suggests a weak positive relationship, though the points remain fairly dispersed.

We also created a county-level income map to provide geographic context for income differences across Washington State.

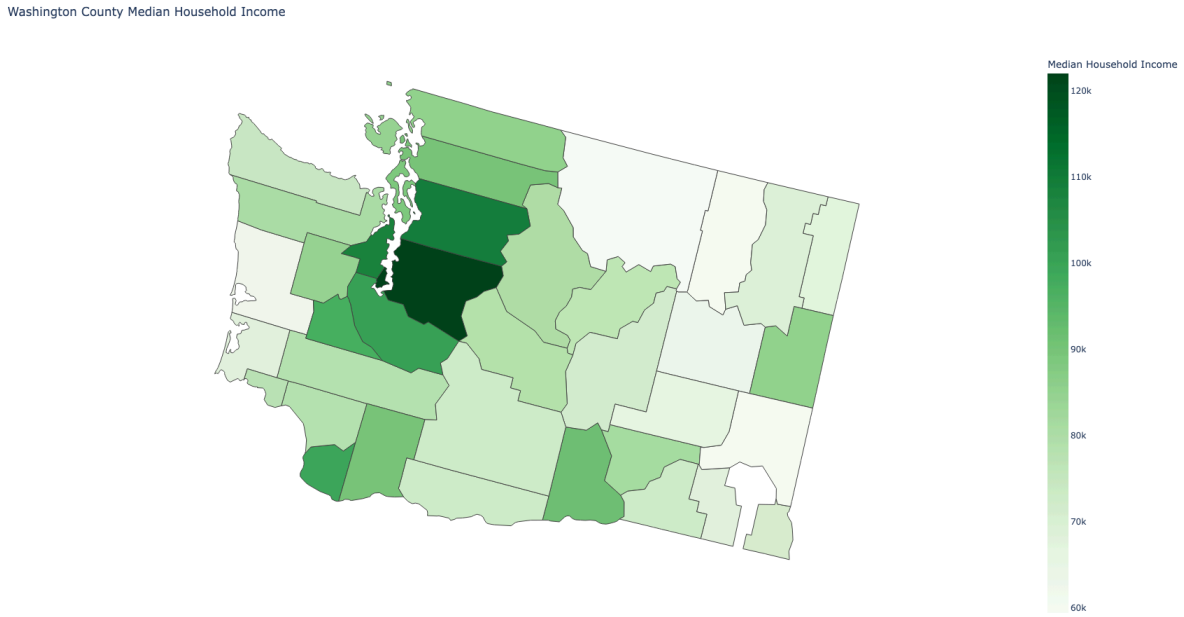

Figure 5. Geographic distribution of median household income across Washington counties. Income levels vary substantially across the state, with some western counties showing relatively higher values.

The scatterplot showed a **weak positive relationship** between median household income and graduation rate. On average, counties with higher median household income tended to have somewhat higher graduation rates. However, the relationship was not strong, and the points were fairly spread out. Several counties had graduation rates that were either higher or lower than would be expected based on income alone, indicating that income is not the only factor associated with educational outcomes.

The geographic income map helped reinforce this interpretation. Some counties with higher median household income were concentrated in western Washington and tended to align with relatively strong graduation outcomes, but this pattern was not perfectly consistent across the state. The map suggests that county wealth may matter, but its effect is uneven and likely interacts with other county-level conditions.

Overall, our results suggest that higher median household income is associated with slightly stronger graduation outcomes, but the relationship is limited in strength. Income appears to be one relevant county-level factor, but not a complete explanation for differences in graduation rate.

6.2 Research Question 2: How does a county's high school graduation rate correlate with the percentage of families in poverty with youth under 18?

To answer this question, we examined the relationship between county graduation rate and child poverty percentage for ages 0–17 using a scatterplot with a fitted regression line.

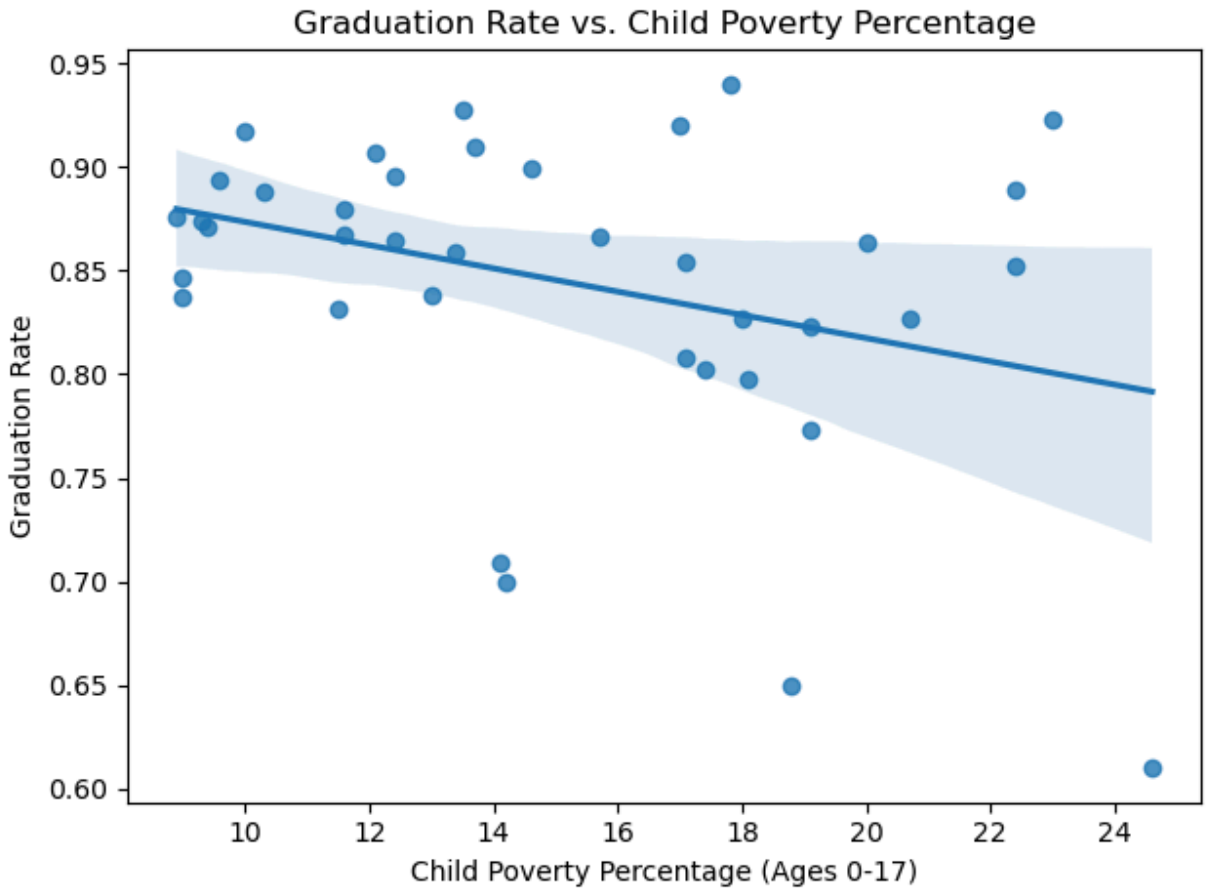

Figure 6. Relationship between county child poverty percentage and graduation rate. The fitted line suggests a negative relationship, with higher child poverty generally corresponding to lower graduation rates.

The scatterplot showed a **negative relationship** between child poverty percentage and graduation rate. In general, counties with higher child poverty percentages tended to have lower graduation rates. Compared with the income relationship in Research Question 1, this downward trend appeared somewhat clearer visually.

At the same time, the relationship was not perfect. There was still noticeable variation among counties, and several counties did not fall closely along the regression line. This indicates that

child poverty percentage is associated with graduation outcomes, but it does not fully determine them.

Even with these limitations, the negative trend supports the broader idea that economic hardship is linked to weaker educational outcomes. Counties with greater poverty among children may face additional barriers related to stability, access to resources, and long-term educational support. Our results therefore suggest that child poverty is a meaningful county-level indicator when studying graduation outcomes in Washington State.

6.3 Research Question 3: Can we predict graduation rates from socioeconomic variables such as income and poverty levels?

To address this question, we built two predictive models using the merged county-level dataset. We used **GraduationRate** as the target variable and used the following socioeconomic indicators as predictors: **MedianHouseholdIncome**, **PovertyPercent_AllAges**, **PovertyEstimate_AllAges**, **ChildPovertyPercent_0_17**, and **ChildPovertyEstimate_0_17**. We compared a **Linear Regression** model and a **Random Forest Regressor** using mean squared error (MSE) and R^2.

The Linear Regression model achieved an **MSE of 0.007339** and an **R^2 of 0.118**. The Random Forest model achieved an **MSE of 0.008208** and an **R^2 of 0.014**. Based on both evaluation metrics, Linear Regression performed better than Random Forest on this dataset.

However, both models showed **limited predictive power overall**, especially as indicated by the low R^2 values. This means that the county-level socioeconomic variables in our dataset explain only a small portion of the variation in graduation rate. Although these variables are related to graduation outcomes, they are not sufficient by themselves to generate strong predictions.

This result is important because it shows that educational outcomes are likely influenced by additional factors not included in our model, such as school-level resources, district policies, teacher availability, demographic composition, transportation access, or other social conditions. In other words, county-level income and poverty measures help reveal broad patterns, but they do not fully capture the complexity of graduation outcomes.

Model	MSE	R^2
Linear Regression	0.007339	0.118
Random Forest	0.008208	0.014

Table 1. Performance comparison of the two predictive models for graduation rate prediction.

7. Impact and Limitations

This project has several practical implications. First, it suggests that county-level economic conditions are meaningfully associated with educational outcomes in Washington State. In particular, higher child poverty is linked to lower graduation rates, and higher median household income is linked to somewhat higher graduation rates. These findings may be useful for policymakers, educators, and community organizations that are interested in targeting resources toward counties facing greater economic hardship.

At the same time, the analysis also shows that these economic indicators provide only limited predictive power. This is important because it prevents overinterpretation. County-level income and poverty measures can help identify broad structural patterns, but they should not be used alone to make strong claims about individual students, schools, or districts.

There are also several limitations to this project. First, the final merged dataset is small, containing only county-level observations for Washington State. This limits the strength of statistical and predictive conclusions. Second, the analysis is based on county-level aggregation, so it cannot capture differences within counties or explain individual student outcomes. Third, the variables in the merged dataset focus on income and poverty, but graduation rates are likely affected by many other factors not included here, such as school funding, program availability, staffing, demographics, and local policy decisions. Finally, the relationships we observe are correlational, not causal. A county with higher poverty may also differ from other counties in many other ways, so the analysis should not be interpreted as proving that one variable directly causes another.

Because of these limitations, our conclusions should be used as evidence of broad county-level association rather than precise prediction or causal explanation. The project is most useful as a reproducible exploratory and modeling framework that highlights where economic environment appears to matter and where additional variables are needed.

8. Challenge Goals

This project addresses two challenge goals: **Machine Learning** and **New Library**.

- ## Machine Learning

  We satisfied the Machine Learning challenge goal by building and comparing two predictive models for graduation rate: a **Linear Regression** model and a **Random Forest Regressor**. These models used multiple socioeconomic variables as predictors and were evaluated using **MSE** and **R^2**. By comparing model performance rather than using only a single model, we were able to assess which approach worked better for this dataset. Our results showed that Linear Regression outperformed Random Forest, although both models had limited predictive power overall.

- **New Library**

  We satisfied the New Library challenge goal by using **Plotly** to create interactive county-level choropleth maps of graduation rate and median household income in Washington State. These visualizations added a geographic dimension to the analysis that was not captured by static plots alone. In particular, the maps helped reveal spatial variation across counties and supported interpretation of the education and income patterns in the dataset.

9. Work Plan Evaluation

Our original work plan was generally accurate in identifying the main stages of the project, but some tasks required more adjustment than expected.

The early data-cleaning and dataset-integration tasks were close to our original expectations. Filtering the OSPI dataset to county-level observations and identifying the correct variables from the SAIPE dataset were manageable, although cleaning and aligning the datasets still took careful attention because the original OSPI data included many rows, multiple cohort definitions, and substantial missingness.

The exploratory data analysis stage also followed our original plan, but it expanded somewhat as we refined which plots were most useful. In addition to the original static visualizations, we added county-level geographic visualizations using Plotly, which improved the final project but required additional setup and formatting work.

The predictive modeling stage also changed somewhat from our initial expectations. While we were able to implement both Linear Regression and Random Forest models, the results showed that the available county-level socioeconomic variables were not especially strong predictors of graduation rate. This meant that a large part of the work became interpretation and limitation analysis rather than purely model optimization.

Overall, our work plan successfully guided the structure of the project, but the later stages required more time for interpretation, visualization refinement, and report writing than we initially anticipated.

10. Testing

We tested our code at multiple stages to make sure the results in this report were trustworthy.

First, we tested the dataset merging process by checking the structure of the merged dataset. We verified that the merged dataset had the expected number of rows and columns, that each county appeared only once, and that the key variables were present after merging. We also

checked that the numeric variables were successfully converted to numeric type and that the value ranges were reasonable for graduation rate, income, and poverty variables.

Second, we used summary statistics and visual inspection as part of our validation process. For example, we confirmed that the merged dataset contained realistic graduation rates and county income values before generating plots or running models. We also inspected the generated plots to confirm that the axes, distributions, and regression relationships matched the expected variables.

Third, we used a separate testing file to automate checks on the merged dataset. These tests included assertions about row counts, county uniqueness, missingness, and variable ranges. This helped us catch mistakes in data preparation before moving on to visualization and modeling.

Finally, we compared model results using standard evaluation metrics rather than relying on visual impressions alone. Reporting MSE and R^2 for both models helped us evaluate performance more carefully and supported our conclusion that the predictive strength of the county-level socioeconomic variables was limited.

Together, these testing steps gave us greater confidence that the analysis pipeline, figures, and modeling results were working as intended.

11. Citations and Collaboration

Citations

- U.S. Census Bureau. *SAIPE State and County Estimates for 2024.*
- Washington Office of Superintendent of Public Instruction. *Report Card Graduation 2024–25.*
- Plotly. *Choropleth mapping documentation.*
- scikit-learn. *LinearRegression and RandomForestRegressor documentation.*

Collaboration

We worked together as project partners on dataset selection, data cleaning, analysis design, visualization, and report writing. We also referred to course materials, official library documentation, and assignment specifications while developing the project.